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Research article

A novel dynamic outlier-robust Kalman filter with Moving Horizon Estimation

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ABSTRACT

The existence of dynamic outliers poses a significant challenge to the Kalman filter (KF). In addressing this challenge, this paper presents an innovative solution: Firstly, by analyzing a period of measurement information to more accurately identify state and measurement dynamic outliers, the system's capacity to adapt to dynamic changes is significantly improved. Next, noise is modeled as a Gaussian-Student's t mixture distribution (GSTM), with mixed model parameters inferred using the variational Bayesian (VB) method based on measurement information, cleverly integrated into the Moving Horizon Estimation (MHE) framework, thus enhancing the flexibility and accuracy of the noise model. Lastly, the optimal window size was identified through simulation experiment analysis to further increase the estimation accuracy. Simulation results demonstrate that the proposed filter exhibits stronger robustness in resisting dynamic outliers compared to existing filters.

1. Introduction

The KF finds extensive application in the field of automatic control, information fusion, and signal processing [1–3]. The KF is fundamentally a minimum variance online recursive estimator, capable of achieving optimal estimation in linear Gaussian systems. However, in certain practical scenarios, such as navigation systems where GPS signals may be obstructed or disturbed due to multipath effects [4], target tracking systems may encounter sensor errors or environmental disturbances [5], all can lead to unknown outliers. These outliers lead to a heavy-tailed distribution of state and measurement noise, which contravenes the assumption in the KF that noise is precisely Gaussian, and thus compromises the accuracy and reliability of the filter [6–9].

To address the challenges mentioned above, some robust filters are proposed [10–16]. While the Huber-based KF (HKF) [17] and the maximum correntropy KF (MCKF) [14] mitigating the influence of outliers to a certain extent, they fall short of fully leveraging the statistical characteristics of heavy-tailed noise, leading to a reduction in estimation accuracy. As research progresses, scholars have identified that precise selection of the noise model correlates positively with enhanced estimation accuracy. Numerous filters have been proposed to model heavy-tailed noise through the utilization of the Student's t distribution [6,18,19]. However, Student's t filters demonstrate limited estimation accuracy when faced with moderately heavy-tailed noises. To address the design shortcomings of Student's t filters, Huang et al. [20] introduced the robust Student's t-based Kalman filter (RSTKF). The RSTKF employs a Student's t distribution to model noise, and adaptively estimates unknown parameters through the VB method. To more effectively distinguish state and measurement outliers, Zhu et al. [21] introduced a sliding window variational outlier-robust KF (SWRKF). However, influenced by the environment, outliers may exhibit dynamic characteristics. To address this, Huang et al. [22] proposed a robust GSTM distribution based KF (GSTMKF) to handle non-stationary heavy-tailed noises. The filter takes into account that noise is subject to different distributions in different time periods.

The GSTMKF resists state and measurement dynamic outliers by utilizing measurements from the current step. However, relying solely on current measurement information is insufficient to accurately distinguish between the two sorts of dynamic outliers. This challenge stems from the complexity of dynamic systems, where noise characteristics and outlier patterns may evolve over time. Inspired by this, this paper proposes a novel dynamic outlier-robust KF with MHE. By leveraging multiple measurements within a sliding window to better identify state and measurement dynamic outliers. The proposed filter's advantages are verified through numerical simulation and target tracking. Taxonomy Table 1 compares and summarizes the existing robust filters. This paper's contributions are 3-aspects:

- An innovative MHE robust KF algorithm is introduced. This method more effectively distinguishes between state and measurement dynamic outliers by analyzing measurement data over a period of time, enhancing the system's adaptability to dynamic

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Table 1
Taxonomy of robust filtering approaches.

Categories	Filters	Main Ideas	Advantages	Shortcomings Compared to the Proposed Filter
M-estimation	HKF [17]	Minimizing the combination of L1 and L2 norm.	Mitigated the adverse effects of heavy-tailed noise to a certain degree.	(1) Difficult to differentiate between the two types of outliers. (2) Failed to fully leverage the inherent heavy-tailed characteristics of noise. (3)Performs poorly under dynamic outliers.
	MCKF [14]	Maximum correntropy criterion		
Student's t filters	STF [19]	Student's t modeling	Improve estimation accuracy by modeling noise.	(1) Difficult to differentiate between the two types of outliers. (2) The assumption that noise parameters are set to fixed values is difficult to satisfy in practical applications. (3) Performs poorly under dynamic outliers.
VB-based KFs	RSTKF [20]	Student's t modeling and VB method	Adaptive estimation of model parameters	(1) Difficult to differentiate between the two types of outliers. (2) Performs poorly under dynamic outliers.
	SWRKF [21]	(1) Student's t modeling and VB method. (2) Silding window	Better identification of outliers.	Performs poorly under dynamic outliers.
	GSTM-RKF [22]	GSTM Distribution Modeling and VB method.	Enable handling of dynamic outliers.	Difficult to differentiate between the two types of dynamic outliers.
	Proposed filter	(1) GSTM Distribution modeling and VB method. (2)MHE	(1) Better identification of dynamic outliers. (2) improve estimation accuracy.	

changes and the accuracy of estimations. Compared to existing technologies, this algorithm not only improves the handling of dynamic outliers but also increases the efficiency of data utilization.

- State and measurement noises are modeled as GSTM distributions, with the parameters of the mixed model adaptively learned through the VB method and integrated into the proposed MHE framework. This combination of mixed distribution modeling and adaptive learning mechanisms not only enhances the flexibility and accuracy of noise modeling but also significantly improves the system's adaptability to complex noise environments and overall performance.
- Simulations confirmed that the proposed filter performs better in handling dynamic outliers compared to traditional robust filters. Additionally, experimental analysis determined the optimal window length, which has improved the accuracy of estimations and enhanced the proposed filter's adaptability to dynamic outliers.

2. Problem formulation

Considering a linear system subjected to non-stationary heavy-tailed noises:

$$X_t = F_{t,t-1}X_{t-1} + w_{t-1} \tag{1}$$

$$Z_t = H_t x_t + v_t \tag{2}$$

where $X_t \in R^n$ represents the state, $Z_t \in R^m$ denotes the measurement information, $F_{t,t-1}$ and H_t indicate the state transition and measurement model matrices, w_t and v_t are the non-stationary heavy-tailed noises.

The update of KF is divided into five steps, namely:

$$\hat{X}_{t|t-1} = F_{t,t-1}\hat{X}_{t-1|t-1} \tag{3}$$

$$P_{t|t-1} = F_{t,t-1}P_{t-1}F_{t,t-1}^{\mathrm{T}} + Q_{t-1} \tag{4}$$

$$\hat{X}_t = \hat{X}_{t|t-1} + K_t(Z_t - H_t\hat{X}_{t|t-1}) \tag{5}$$

$$K_t = P_{t|t-1}H_t^{\mathrm{T}}(H_tP_{t|t-1}H_t^{\mathrm{T}} + R_t)^{-1} \tag{6}$$

$$P_t = (I_n - K_tH_t)P_{t|t-1} \tag{7}$$

In Eqs. (3) and (4), the time updates of the KF are presented, where $\hat{X}_{t|t-1}$ and $P_{t|t-1}$ denote the prediction state and the covariance matrix of prediction errors, respectively. Eqs. (5) to (7) correspond to the measurement updates of the KF, with $\hat{X}_t$ and P_t representing the posterior state estimate and posterior estimation error covariance matrix, respectively. The Kalman Filter gain is denoted by K_t. The Kalman Smoother (KS) involves the following four steps:

$$P_{s,t|t-1} = F_{t,t-1}P_{t-1}F_{t,t-1}^{\mathrm{T}} + Q_t \tag{8}$$

$$G_{t-1} = P_{t-1}F_{t,t-1}^{\mathrm{T}}P_{s,t|t-1}^{-1} \tag{9}$$

$$\hat{X}_{t-1|T} = \hat{X}_{t-1} + G_{t-1}(\hat{X}_{t|T} - F_{t,t-1}\hat{x}_{t-1}) \tag{10}$$

$$P_{t-1|T} = P_{t-1} + G_{t-1}(P_{t|T} - P_{s,t|t-1})G_{t-1}^{\mathrm{T}} \tag{11}$$

where, G_{t-1} represents the smoothing gain matrix, which serves as the correction applied to the state estimate during the smoothing step. $\hat{X}_{t-1|T}$ and $P_{t-1|T}$ refer to the covariance matrices associated with the final smoothed state estimate and its uncertainty, respectively. However, relying solely on measurement information from the current step is insufficient for distinguishing between state and measurement dynamic outliers, potentially leading to error accumulation over time.

Motivated by this consideration, this paper proposes a novel dynamic outlier-robust KF integrated with MHE.

3. Proposed algorithm

3.1. Moving horizon estimation

MHE is a model predictive control method employed for the estimation of system states [23,24]. By introducing a sliding window with a length of M, the problem of state estimation is reformulated within an optimization framework. Update state estimates by minimizing the cost function [25–27].

$$\phi_t^* = \min_{x_{t-M},\{w_t\}_{j=t-M}^{t-1}} \phi_k(x_{t-M}, \{w_t\}_{j=t-M}^{t-1}) \tag{12}$$

where, the constraint condition is:

$$x_j \in X for j = (t-M), \ldots, T, w_j \in W for j = (t-M), \ldots, (t-1). \tag{13}$$

the objective function expression is as follows:

$$\phi_t(x_{t-L}, \{w_t\}_{j=t-M}^{t-1}) = \sum_{j=t-M}^{t-1} v_j^{\mathrm{T}} R^{-1} v_j + w_j^{\mathrm{T}} Q^{-1} w_j + \phi_{t-M}(x_0, \{w_t\}_{j=0}^{t-M-1}) \tag{14}$$

where $\{w_t\}_{j=t-M}^{t-1}$ and $\{w_t\}_{j=0}^{t-M-1}$ are the sequence of disturbing variables. The notion of arrival cost is introduced, playing a crucial role in guaranteeing the approximation of the estimated final state to the system's target state.

In general, the efficiency of MHE is contingent upon the selection of the estimated window length M, which cannot increase indefinitely because of the limited onboard computing power. In addition, the estimation accuracy does not always increase with M, because model mismatches can reduce the effectiveness of model predictions, which can adversely affect the estimation performance.

3.2. Noise modeling

In actual engineering scenarios, some non-Gaussian noises caused by dynamic outliers result in non-stationary heavy-tailed noise. This paper models the noise as a GSTM distribution.

The random vector and the mixed probability are respectively assumed to be A and B, with the probability density function (PDF) of A under the GSTM distribution, conditioned on B, being formulated as

$$P(A|B) = BN(A; \upsilon, \Sigma) + (1-B)St(A; \upsilon, \Sigma, \iota) \tag{15}$$

where, υ represents the mean, while Σ and ι correspond to the scale matrix and dof parameter, respectively. Additionally, B is presumed to be distributed according to a Beta distribution [22]

$$P(B) = Be(B; \phi, 1-\phi) \tag{16}$$

where, ϕ is the prior mixing probability, $E[B] = \int Be(B; \phi, 1-\phi)B = \phi$. Based on Eqs. (15)–(16), the PDF of the random vector A, which follows a GSTM distribution, can be expressed as

$$P(A) = \int P(A|B)P(B)dB = \int Be(B; \phi, 1-\phi)BdB\,N(A; \upsilon, \Sigma) + \int Be(B; \phi, 1-\phi)(1-B)dB\,St(A; \upsilon, \Sigma, \iota) \tag{17}$$

Using (17), we have

$$P(A) = \phi N(A; \upsilon, \Sigma) + (1-\phi)St(A; \upsilon, \Sigma, \iota) \tag{18}$$

From Eq. (18), it becomes evident that by choosing the prior mixing probability ϕ as 1 or 0, the GSTM distribution reverts to the Gaussian distribution and Student's t distribution, respectively.

The transition and likelihood PDF within the time interval $s \in [t-M+1, t]$ are represented using GSTM distributions.

$$P(X_s|X_{s-1}, \alpha_s, \eta_s) = \alpha_s N(X_s; F_s X_{s-1}, Q_t) + (1-\alpha_s)N(X_s; F_s X_{s-1}, Q_t/\eta_s) \tag{19}$$

$$P(Z_s|H_s X_s, \beta_s, \delta_s) = \beta_s N(Z_s; H_s X_s, R_t) + (1-\beta_s)N(Z_s; H_s X_s, R_t/\delta_s) \tag{20}$$

$$P(\eta_s) = G(\eta_s; \frac{a}{2}, \frac{a}{2}) \tag{21}$$

$$P(\delta_s) = G(\delta_s; \frac{b}{2}, \frac{b}{2}) \tag{22}$$

where $N(\cdot; l, L)$ is a Gaussian distribution, l and L represent the mean and variance, respectively. $G(\cdot; c, d)$ is a Gamma distribution, where c and d denote the shape and rate parameters, respectively [28]. α_s and β_s are the mixing probabilities for Gaussian and Student's t distributions. The aforementioned mixing probabilities are modeled by $Be(\cdot; g, h)$, where g and h denote the prior shape parameters.

$$P(\alpha_s) = Be(\alpha_s; e_0, 1-e_0) \tag{23}$$

$$P(\beta_s) = Be(\beta_s; f_0, 1-f_0) \tag{24}$$

The mixing probabilities α_s and β_s in the GSTM distribution are based on the unknown premise and need to be inferred jointly by the VB method. Here Bernoulli random variables j and k are introduced as conjugate prior distributions of mixed probabilities α_s and β_s

$$P(X_s|X_{s-1}, \eta_s, j_s) = [N(X_s; F_s X_{s-1}, Q_t)]^{\mathrm{j_s}} \times [N(X_s; F_s X_{s-1}, Q_t/\eta_s)]^{1-\mathrm{j_s}} \tag{25}$$

$$P(Z_s|H_s X_s, \delta_s, k_s) = [N(Z_s; H_s X_s, R_t)]^{\mathrm{k_s}} \times [N(Z_s; H_s X_s, R_t/\delta_s)]^{1-\mathrm{k_s}} \tag{26}$$

$$P(j_s|\alpha_s) = {\alpha_s}^{\mathrm{j_s}}(1-\alpha_s)^{1-\mathrm{j_s}} \tag{27}$$

$$P(k_s|\beta_s) = {\beta_s}^{\mathrm{k_s}}(1-\beta_s)^{1-\mathrm{k_s}} \tag{28}$$

by the above formula in $s \in [t-M+1, t]$, a novel dynamic outlier-robust KF with MHE is derived through the VB method.

Remark 1. Fig. 1 shows the density comparison of Gaussian (G), Student's t (ST), and GSTM distributions. Their respective PDFs are defined as $P_G(A) = N(A; \upsilon, \Sigma)$, $P_{ST}(A) = St(A; \upsilon, \Sigma, \iota)$, and $P_{GSTM}(A) = \phi P_G(A) + (1-\phi)P_{ST}(A)$, where $\upsilon = 0$, $\Sigma = 1$, $\iota = 1$ and $\phi = 0.5$. The figure clearly shows that the tail of the GSTM distribution is heavier than that of the Gaussian distribution, yet lighter than that of the ST distribution. The uniqueness of the GSTM distribution lies in its degeneration to the Gaussian distribution in the absence of outliers, and to the ST distribution in the presence of outliers. This characteristic of the GSTM distribution is primarily determined by the mixture probability. In the proposed filtering approach, the mixture probability employs a VB-based adaptive learning mechanism, dynamically adjusting between the Gaussian and ST distributions according to the actual noise conditions. This suggests that the GSTM distribution is particularly well-suited for handling non-stationary heavy-tailed noise caused by dynamic outliers.

3.3. Variational Bayesian MHE-based algorithm

Because full information estimation (FIE) depends on the whole sequencing column $Z_{1:t}$, it will cause difficulties such as error accumulation and computation burden [29]. To enhance estimation accuracy, MHE with a fixed window length denoted as M is employed for variational inference within the time series $s \in [t-M+1, t]$. In this section, the VB method is utilized to achieve the optimal approximation of the joint posterior PDF $P(\Xi_t|Z_{1:t})$[30], where $\Xi_t = \{X_{t-M:t}, \eta_s, \delta_s, j_s, k_s, \alpha_s, \beta_s\}$.

$$P(\Xi_t|Z_{1:k}) \approx q(X_{t-M:t})q(\eta_s)q(\delta_s)q(j_s)q(k_s)q(\alpha_s)q(\beta_s) \tag{29}$$

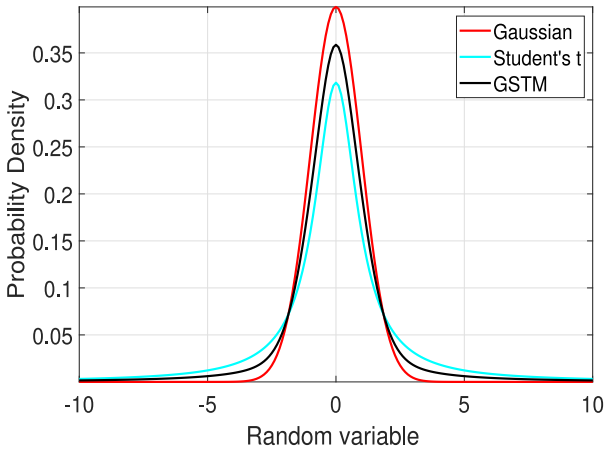


Fig. 1. Density of different distributions.

$$logq(\theta_t) = E_{(\Xi_t-\theta_t)}[logp(\Xi_t, z_{1:t})] + C_\theta \tag{30}$$

where θ_t represents an element in Ξ_t and C_θ represents a constant that does not contain θ_t. Applying Bayes' theorem, the joint PDF can be articulated in a factorized representation as follows:

$$\begin{aligned} P(\Xi_t, Z_{1:t}) = & \prod_{s=t-M+1}^{t} [[N(X_s; F_s X_{s-1}, Q_t)]^{j_s} \\ & \times [N(X_s; F_s X_{s-1}, Q_t/\eta_s)]^{1-j_s} [N(Z_s; H_s X_s, R_t)]^{k_s} \\ & \times [N(Z_s; H_s X_s, R_t/\delta_s)]^{1-k_s} G(\eta_s; \frac{a}{2}, \frac{a}{2}) G(\delta_s; \frac{b}{2}, \frac{b}{2}) \\ & \times Be(\alpha_s; e_0, 1-e_0) Be(\beta_s; f_0, 1-f_0) \alpha_s{}^{j_s} (1-\alpha_s)^{1-j_s} \\ & \times \beta_s{}^{k_s} (1-\beta_s)^{1-k_s}] N(X_{t-M}, \hat{X}_{t-M|t-M}, P_{t-M|t-M}) \end{aligned} \tag{31}$$

Proposition 1. *Setting $\theta_t = X_{t-M:t}$ in (30), the initial distribution is given as*

$$P(X_{t-M}|Z_{1:t-M}) = N(X_{t-M}; \hat{X}_{t-M|t-M}, P_{t-M|t-M}) \tag{32}$$

the scale matrices $\hat{Q}_{s|t}^{(l)}$ and $\hat{R}_{s|t}^{(l)}$ are updated as

$$\hat{Q}_{s|t}^{(l)} = \frac{Q_t}{E^{(l)}[j_s] + (1 - E^{(l)}[j_s]) E^{(l)}[\eta_s]} \tag{33}$$

$$\hat{R}_{s|t}^{(l)} = \frac{R_t}{E^{(l)}[k_s] + (1 - E^{(l)}[k_s]) E^{(l)}[\delta_s]} \tag{34}$$

Proof. See Appendix A.1.

Proposition 2. *Let $\theta_t = \eta_s$ and $\theta_t = \delta_s$, and putting (31) in (30) has*

$$\begin{aligned} logq^{(l+1)}(\eta_s) = & \sum_{s=t-M+1}^{t} [0.5a + 0.5n(1 - E^{(l+1)}[j_s]) - 1] log\eta_s \\ & - 0.5[a + (1 - E^{(l)}[j_s]) tr(A_s{}^{(l+1)} Q_t)] \eta_s + C_\eta \end{aligned} \tag{35}$$

$$\begin{aligned} logq^{(l+1)}(\delta_s) = & \sum_{s=t-M+1}^{t} [0.5b + 0.5m(1 - E^{(l+1)}[k_s]) - 1] log\delta_s \\ & - 0.5[b + (1 - E^{(l)}[k_s]) tr(B_s{}^{(l+1)} R_t)] \delta_s + C_\delta \end{aligned} \tag{36}$$

and according to (35) and (36), the result can be obtained:

$$q^{(l+1)}(\eta_s) = G(\eta_s; \hat{o}_{s|t}^{(l+1)}, \hat{p}_{s|t}^{(l+1)}) \tag{37}$$

$$q^{(l+1)}(\delta_s) = G(\delta_s; \hat{q}_{s|t}^{(l+1)}, \hat{r}_{s|t}^{(l+1)}) \tag{38}$$

where the shape parameters $\hat{o}_{s|t}^{(l+1)}$ and $\hat{p}_{s|t}^{(l+1)}$, the rate parameters $\hat{q}_{s|t}^{(l+1)}$ and $\hat{r}_{s|t}^{(l+1)}$ are given by

$$\hat{o}_{s|t}^{(l+1)} = 0.5a + 0.5n(1 - E^{(l+1)}[j_s]) \tag{39}$$

$$\hat{p}_{s|t}^{(l+1)} = 0.5[a + (1 - E^{(l+1)}[j_s]) tr(A_s{}^{(l+1)} Q_t{}^{-1})] \tag{40}$$

$$\hat{q}_{s|t}^{(l+1)} = 0.5b + 0.5n(1 - E^{(l+1)}[k_s]) \tag{41}$$

$$\hat{r}_{s|t}^{(l+1)} = 0.5[b + (1 - E^{(l+1)}[k_s]) tr(B_s{}^{(l+1)} R_t{}^{-1})] \tag{42}$$

and the expressions for the auxiliary matrix $A_s{}^{(l+1)}$ and $B_s{}^{(l+1)}$ mentioned above are

$$A_s^{(l+1)} = E^{(l+1)}[(X_s - F_s X_{s-1})(X_s - F_s X_{s-1})^T] \tag{43}$$

$$B_s^{(l+1)} = E^{(l+1)}[(z_s - H_s X_s)(z_s - H_s X_s)^T] \tag{44}$$

Proof. See Appendix A.2.

Proposition 3. *Let $\theta_t = j_s$ and $\theta_t = k_s$, and substituting (31) in (30) gets*

$$\begin{aligned} logq^{(l+1)}(j_s) = & \sum_{s=t-M+1}^{t} [j_s E^{(l)}[log\alpha_s] - 0.5 tr(A_s{}^{(l+1)} Q_t{}^{-1}) \\ & + (1 - j_s)(E^{(l)}[log(1 - \alpha_s)]) + 0.5n E^{(l+1)}[log\eta_s] \\ & - 0.5 E^{(l+1)}[\eta_s] tr(A_s{}^{(l+1)} Q_t{}^{-1})] + C_j \end{aligned} \tag{45}$$

$$\begin{aligned} logq^{(l+1)}(k_s) = & \sum_{s=t-M+1}^{t} [k_s E^{(l)}[log\beta_s] - 0.5 tr(B_s{}^{(l+1)} R_t{}^{-1}) \\ & + (1 - k_s) E^{(l)}[log(1 - \beta_s)] + 0.5m E^{(l+1)}[log\delta_s] \\ & - 0.5 E^{(l+1)}[\delta_s] tr(B_s{}^{(l+1)} R_t{}^{-1})] + C_k \end{aligned} \tag{46}$$

$logq^{(l+1)}(j_s)$ and $logq^{(l+1)}(k_s)$ are approximately Bernoulli distributions. When j_s and k_s take probabilities one and unity respectively, they are denoted as

$$\begin{aligned} P(j_s = 0) = & r_j exp(E^{(l)}[log(1 - \alpha_s)] + 0.5n E^{(l+1)}[log\eta_s] \\ & - 0.5 E^{(l+1)}[\eta_s] tr(A_s{}^{(l+1)} Q_t{}^{-1})) \end{aligned} \tag{47}$$

$$P(j_s = 1) = r_j exp(E^{(l)}[log\alpha_s] - 0.5 tr(A_s{}^{(l+1)} Q_t{}^{-1})) \tag{48}$$

$$\begin{aligned} P(k_s = 0) = & r_k exp(E^{(l)}[log(1 - \beta_s)] + 0.5m E^{(l+1)}[log\delta_s] \\ & - 0.5 E^{(l+1)}[\delta_s] tr(B_s{}^{(l+1)} R_t{}^{-1})) \end{aligned} \tag{49}$$

$$P(k_s = 1) = r_k exp(E^{(l)}[log\beta_s] - 0.5 tr(B_s{}^{(l+1)} R_t{}^{-1})) \tag{50}$$

where r_j and r_k are normalizing constants.

Proof. See Appendix A.3.

Proposition 4. *Let $\theta_t = \alpha_s$ and $\theta_t = \beta_s$, and putting (31) in (30) arrives*

$$\begin{aligned} logq^{(l+1)}(\alpha_s) = & \sum_{s=t-M+1}^{t} [(E^{(l+1)}[j_s] + e_0 - 1) log\alpha_s \\ & + (1 - E^{(l+1)}[j_s] - e_0) log(1 - \alpha_j)] + C_\alpha \end{aligned} \tag{51}$$

$$\begin{aligned} logq^{(l+1)}(\beta_s) = & \sum_{s=t-M+1}^{t} [(E^{(l+1)}[k_s] + f_0 - 1) log\beta_s \\ & + (1 - E^{(l+1)}[k_s] - f_0) log(1 - \beta_j)] + C_\beta \end{aligned} \tag{52}$$

update $logq^{(l+1)}(\alpha_s)$ and $logq^{(l+1)}(\beta_s)$ to Beta distribution. $\hat{u}_{s|t}^{(l+1)}$, $\hat{y}_{s|t}^{(l+1)}$, $\hat{\mu}_{s|t}^{(l+1)}$, and $\hat{v}_{s|t}^{(l+1)}$ are shape parameters.

$$q^{(l+1)}(\alpha_s) = Be(\alpha_s; \hat{u}_{s|t}^{(l+1)}, \hat{y}_{s|t}^{(l+1)}) \tag{53}$$

$$q^{(l+1)}(\beta_s) = Be(\beta_s; \hat{\mu}_{s|t}^{(l+1)}, \hat{v}_{s|t}^{(l+1)}) \tag{54}$$

$$\hat{u}_{s|t}^{(l+1)} = e_0 + E^{(l+1)}[j_s] \tag{55}$$

$$\hat{y}_{s|t}^{(l+1)} = 2 - e_0 - E^{(l+1)}[j_s] \tag{56}$$

$$\hat{\mu}_{s|t}^{(l+1)} = f_0 + E^{(l+1)}[k_s] \tag{57}$$

$$\hat{v}_{s|t}^{(l+1)} = 2 - f_0 - E^{(l+1)}[k_s] \tag{58}$$

Proof. See Appendix A.4.

The following summarizes the expectations of VB-based on MHE:

$$E^{(l+1)}[\eta_s] = \frac{\hat{o}_{s|t}^{(l+1)}}{\hat{p}_{s|t}^{(l+1)}}, E^{(l+1)}[\delta_s] = \frac{\hat{q}_{s|t}^{(l+1)}}{\hat{r}_{s|t}^{(l+1)}} \tag{59}$$

$$E^{(l+1)}[j_s] = \frac{P(j_s = 1)}{P(j_s = 1) + P(j_s = 0)} \tag{60}$$

$$E^{(l+1)}[k_s] = \frac{P(k_s = 1)}{P(k_s = 1) + P(k_s = 0)} \tag{61}$$

$$E^{(l+1)}[log\eta_s] = \psi(\hat{o}_{s|t}^{(l+1)}) - log\hat{p}_{s|t}^{(l+1)} \tag{62}$$

$$E^{(l+1)}[log\delta_s] = \psi(\hat{q}_{s|t}^{(l+1)}) - log\hat{r}_{s|t}^{(l+1)} \tag{63}$$

$$E^{(l+1)}[log\alpha_s] = \psi(\hat{u}_{s|t}^{(l+1)}) - \psi(2) \tag{64}$$

$$E^{(l+1)}[log(1-\alpha_s)] = \psi(\hat{y}_{s|t}^{(l+1)}) - \psi(2) \tag{65}$$

$$E^{(l+1)}[log\beta_s] = \psi(\hat{\mu}_{s|t}^{(l+1)}) - \psi(2) \tag{66}$$

$$E^{(l+1)}[log(1-\beta_s)] = \psi(\hat{v}_{s|t}^{(l+1)}) - \psi(2) \tag{67}$$

$$\begin{aligned} A_s^{(l+1)} &= (\hat{X}_{s|t}^{(l+1)} - F_s\hat{X}_{s-1|t}^{(l+1)})(\hat{X}_{s|t}^{(l+1)} - F_s\hat{X}_{s-1|t}^{(l+1)})^{\mathrm{T}} \\ &+ F_j P_{s-1|t}^{(l+1)} F_s^{\mathrm{T}} - [P_{s-1,s|t}^{(l+1)}]^{\mathrm{T}} F_s^{\mathrm{T}} - F_s P_{s-1,s|t}^{(l+1)} + \hat{P}_{s|t}^{(l+1)} \end{aligned} \tag{68}$$

$$B_s^{(l+1)} = H_s P_{s|t}^{(l+1)} H_s^{\mathrm{T}} + (Z_s - H_s\hat{X}_{s|t}^{(l+1)})(Z_s - H_s\hat{X}_{s|t}^{(l+1)})^{\mathrm{T}} \tag{69}$$

where $\hat{X}_{s|t}^{(l+1)}$ represents the smoothing estimation, $P_{s|t}^{(l+1)}$ and $P_{s-1,s|t}^{(l+1)}$ represent the smoothing error covariance matrix and the smoothing error cross-covariance matrix respectively.

$$P_{s-1,s|t}^{(l+1)} = G_{s-1}^{(l+1)} P_{s|t}^{(l+1)} \tag{70}$$

$G_{s-1}^{(l+1)}$ is the smoothing gain.

The proposed filter is summarized by Algorithm 1.

Remark 2. In statistical inference, the VB method leverages prior distributions to approximate posterior distributions, effectively merging prior knowledge with measurements [30]. This enhances its adaptability to noise characteristic changes, thus offering more flexible and robust parameter estimation.

Remark 3. In the initialization phase, our algorithm prepares the state estimation using the data up to $t-1$ because at the onset, the data for the current state t is not yet available. As the filter updates with each new time step, the window "slide" to encompass the newest data up to the current time t, thus including time t in the estimation window for subsequent iterations, as demonstrated in Sections 3.2 and 3.3 with the window $[t-M+1, t]$.

4. Simulation

4.1. Numerical simulation

In this section, the effectiveness and superiority of the proposed robust filter are demonstrated through numerical simulations. In the numerical examples, the state and measurement noises are assumed to be non-stationary heavy-tailed noises caused by dynamic outliers. See (71) and (72) for noise characteristics.

$$w_t \begin{cases} p1N(0,Q) + (1-p1)N(0,100Q) & t \in [100,200] \\ p2N(0,Q) + (1-p2)N(0,100Q) & t \in [200,300] \\ N(0,Q) & othertime \end{cases} \tag{71}$$

Table 2
The parameter settings of different filters.

Different filters	Parameter settings
HKF [17]	Tuning parameter $\gamma = 1.345$ max iteration count $i = 50$
GSTMRKF [22]	DOFs $\varpi = \vartheta = 5$ Prior mixing probabilities $g_0 = e_0 = 0.85$ max iteration count $i = 50$
RSTKF [20]	DOFs $\varpi = \vartheta = 5$ IW distributed scale matrices $\tau_\varpi = \tau_\vartheta = 5$ max iteration count $i = 50$
Proposed filter	DOFs $a = b = 5$ Window length $M = 6$ Prior mixing probabilities $e_0 = f_0 = 0.85$ max iteration count $i = 4$

Table 3
The ARMSEs and SSRT of different filters in numerical simulation.

	TKF	HKF	GSTMRKF	RSTKF	Proposed filter
ARMSE	7.55	14.38	25.64	17.25	10.59
Time(ms)	0.00632	0.00448	0.0992	0.5005	0.2402

$$v_t \begin{cases} p1N(0,R) + (1-p1)N(0,100R) & t \in [100,200] \\ p2N(0,R) + (1-p2)N(0,100R) & t \in [200,300] \\ N(0,R) & othertime \end{cases} \tag{72}$$

where the probabilities $p1 = 0.99$ and $p2 = 0.95$. The nominal noise variances are specified as $Q = 100$ and $R = 100$. Simulation time is set to $400s$, Monte Carlo runs $C_{mc} = 1000$. For the purpose of comparison, the GSTMRKF [7], the HKF [31], the RSTKF [20], and the KF with true noise covariance matrices (TKF) are tested. The parameters for the filters employed in the comparison were chosen based on recommendations found in literature, as explicitly outlined in Table 2.

The accuracy of estimates is compared by root-mean-square error(RMSE):

$$RMSE_X(t) = \sqrt{\sum_{n_{mc}=1}^{C_{mc}} (X_t^{(\mathrm{n_{mc}})} - \hat{X}_t^{(\mathrm{n_{mc}})})(X_t^{(\mathrm{n_{mc}})} - \hat{X}_t^{(\mathrm{n_{mc}})})} \tag{73}$$

where $RMSE_X(t)$ is the RMSE of the state vector. $X_t^{(\mathrm{n_{mc}})}$ and $\hat{X}_t^{(\mathrm{n_{mc}})}$ denote the true and estimated states at the $n_{mc}-th$ Monte Carlo run. Based on 1000 Monte Carlo runs, the RMSEs of different filters are shown in Fig. 2. The simulation is segmented into four stages. The second and third stages, respectively, characterize heavy-tailed noise induced by outliers of varying magnitudes, while the remaining two stages involve Gaussian noise. From Fig. 2, it is clear that in terms of accuracy, excluding the unobtainable TKF, the filter proposed in this paper demonstrates superior performance, achieving the smallest RMSE. The average RMSEs (ARMSEs) and single-step run times (SSRT) of different filters in numerical simulation are presented in Table 3. It clearly shows that the ARMSE of the proposed filter is the closest to the ideal, yet unattainable TKF among the algorithms compared. The proposed filter not only achieves the lowest ARMSE but also maintains a moderate implementation time, effectively balancing performance and efficiency. Therefore, the proposed filter shows superiority in resisting dynamic outliers.

4.2. Target tracking

In this part, the proposed filter's advantages are demonstrated through target tracking simulation. In practical scenarios, the omnipresent reflection effects of radar signals and the non-stationary maneuvering of target objects lead to the presence of non-stationary

Algorithm 1 Proposed filter

Input: $\hat{X}_{0|0}, P_{0|0}, Q_t, R_t, M, e_0, f_0, N, \varepsilon, \{Z_{s+1}\}\ s \in [t-M, t-1]$
$\left\{\hat{\eta}_{s|t-1}, \hat{\delta}_{s|t-1}, \hat{j}_{s|t-1}, \hat{k}_{s|t-1}\right\}\ s \in [t-M+1, t-1]$

1: Initialization:
for s=t-M+1:t-1
$E^{(0)}[\eta_s] = \hat{\eta}_{s|t-1}$, $E^{(0)}[\delta_s] = \hat{\delta}_{s|t-1}$
$E^{(0)}[j_s] = \hat{j}_{s|t-1}$, $E^{(0)}[k_s] = \hat{k}_{s|t-1}$
end for
$E^{(0)}[log\alpha_s] = (\psi(e_0) - \psi(1))E^{(0)}[j_s]$
$E^{(0)}[log\beta_s] = (\psi(f_0) - \psi(1))E^{(0)}[k_s]$
$E^{(0)}[log(1-\alpha_s)] = (\psi(1-e_0) - \psi(1))E^{(0)}[j_s]$
$E^{(0)}[log(1-\beta_s)] = (\psi(1-f_0) - \psi(1))E^{(0)}[k_s]$

2: Forward KF:
$[\hat{X}_{t|t}^{(l-1)}, \hat{P}_{t|t}^{(l-1)}] = KF(\hat{X}_{t-1|t-1}^{(l-2)}, \hat{P}_{t-1|t-1}^{(l-2)}, F_t, H_t, Z_t, \frac{Q_t}{E^{(l)}[j_s]+(1-E^{(l)}[j_s])E^{(l)}[\eta_s]}, \frac{R_t}{E^{(l)}[k_s]+(1-E^{(l)}[k_s])E^{(l)}[\delta_s]})$

3: Backward KS:
for s=t:(-1):t-M+1
$[\hat{X}_{s-1|t}^{(l-1)}, \hat{P}_{s-1|t}^{(l-1)}] = KS(\hat{X}_{s|t}^{(l-1)}, \hat{P}_{s|t}^{(l-1)}, \hat{X}_{s-1|s-1}^{(s-2)}, \hat{P}_{s-1|s-1}^{(s-2)}, F_s, \frac{Q_t}{E^{(l)}[j_s]+(1-E^{(l)}[j_s])E^{(l)}[\eta_s]})$
Calculate A_s and B_s using (68)–(69).
end for

4: Update $q^{(l+1)}(\eta_s)$ and $q^{(l+1)}(\delta_s)$ using (37)–(42).
Calculate j_s and k_s using (47)–(50).
Update $q^{(l+1)}(\alpha_s)$ and $q^{(l+1)}(\beta_s)$ using (53) and (58).

5: if $\|\hat{X}_{t|t}^{(l+1)} - \hat{X}_{t|t}^{(l)}\|/\|\hat{X}_{t|t}^{(l)}\| \le \varepsilon$ break
end for

6: Data saving:
for s=t-M+2:t
$\hat{\eta}_{s|t} = E^{(l+1)}[\eta_s]$, $\hat{\delta}_{s|t} = E^{(l+1)}[\delta_s]$
$\hat{j}_{s|t} = E^{(l+1)}[j_s]$, $\hat{k}_{s|t} = E^{(l+1)}[k_s]$
end for

Output: $\left\{\hat{\eta}_{s|t}, \hat{\delta}_{s|t}, \hat{j}_{s|t}, \hat{k}_{s|t}\right\}\ s \in [t-M+2, t]$, $\left\{\hat{X}_{s|s}, P_{s|s}\right\}\ s \in [t-M+1, t]$.

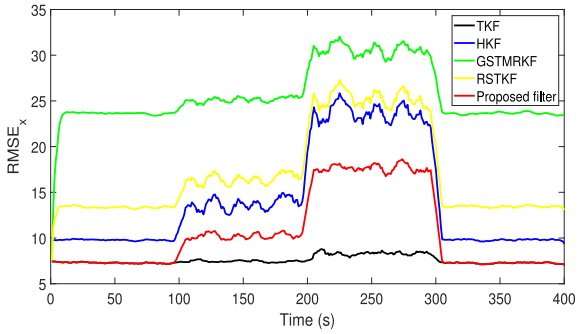


Fig. 2. RMSEs of the state.

heavy-tailed noises. Addressing this prevalent phenomenon becomes crucial, underscoring its significance [28]. The estimated state includes velocity and position, namely, $[x_k, y_k, \dot{x}_k, \dot{y}_k]$. The coefficient matrices are given by [32]:

$$F_{t,t-1} = \begin{bmatrix} I_2 & \triangle t I_2 \\ 0 & I_2 \end{bmatrix}, H_t = [I_2, 0] \tag{74}$$

The attributes of non-stationary heavy-tailed noises are delineated as follows:

$$w_t \begin{cases} p1N(0,Q) + (1-p1)N(0,U1Q) & t \in [100,200] \\ p2N(0,Q) + (1-p2)N(0,U1Q) & t \in [200,300] \\ N(0,Q) & othertime \end{cases} \tag{75}$$

$$v_t \begin{cases} p1N(0,R) + (1-p1)N(0,U2R) & t \in [100,200] \\ p2N(0,R) + (1-p2)N(0,U2R) & t \in [200,300] \\ N(0,R) & othertime \end{cases} \tag{76}$$

where the normal probabilities are set as $p1 = 0.95, p2 = 0.80$. The covariance magnifications for the state and measurement outliers are set at $U1 = 800$ and $U2 = 100$. The Q and R are specified as

$$Q = 0.1 \begin{bmatrix} \frac{\triangle t^3}{3} I_2 & \frac{\triangle t^2}{2} I_2 \\ \frac{\triangle t^2}{2} I_2 & \triangle I_2 \end{bmatrix}, R = 10 I_2 \tag{77}$$

In the simulation, $X_0 = [0,0,0,0]^{\mathrm{T}}, P_0 = diag([10000, 10000, 100, 100])$. For comparison, the TKF, the HKF [31], the GSTMRKF [7], the SWRKF [21], and the RSTKF [20] are tested. Where the DOF of SWRKF is set to 5 and the other parameters are set as window length $M = 10$, IW distributed scale matrices $y_{0|0} = u_{0|0} = 5$. The parameter settings of the TKF, the HKF, the GSTMRKF, the RSTKF, and the proposed filter are consistent with the numerical simulation. The Monte Carlo simulation is conducted for 1000 times, with a simulation duration of 400 s.

The estimated results from all filters are showcased in Figs. 3 and 4. During the time intervals of 0–100 s and 300–400 s, the state and measurement noise are characterized by Gaussian distribution. In the intervals of 100–200 s and 200–300 s, the state and measurement noise exhibits varying degrees of a heavy-tailed distribution. The figures indicates that, aside from the unavailable TKF, the proposed filter achieves the best RMSE performance in position and velocity. This suggests that the proposed filter can more accurately represent non-stationary heavy-tailed noise through adaptive learning of mixed probabilities. The ARMSEs and SSRT of different filters in target tracking are detailed in Table 4. It is evident that the proposed filter achieves the smallest ARMSE for velocity and position compared to the other filters evaluated, and is closest to TKF. Analysis of the SSRT in the table reveals that the computation time of the proposed filter is less than that of the SWRKF, although it is slightly higher than that of other filters. This suggests that the proposed filter balances moderate computational efficiency with superior estimation accuracy compared to existing filters.

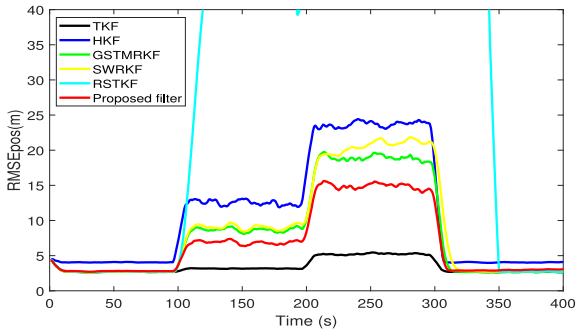


Fig. 3. RMSE of position.

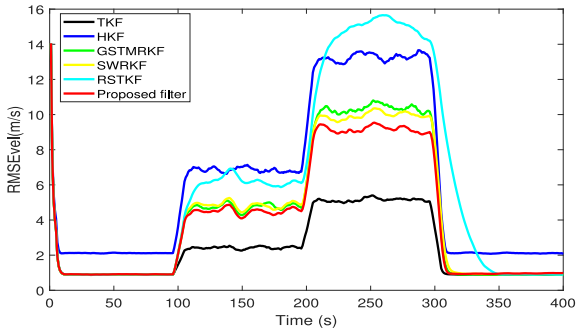


Fig. 4. RMSE of velocity.

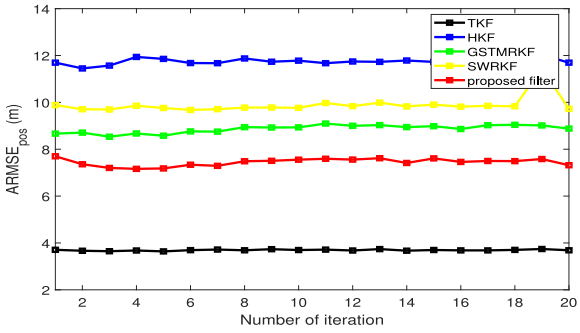


Fig. 5. ARMSE of position when $i = 1, 2, 3, \ldots, 20$.

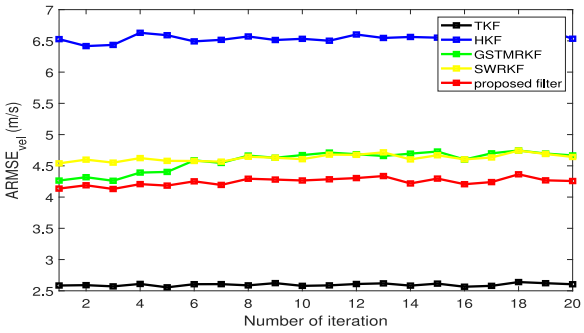


Fig. 6. ARMSE of velocity when $i = 1, 2, 3, \ldots, 20$.

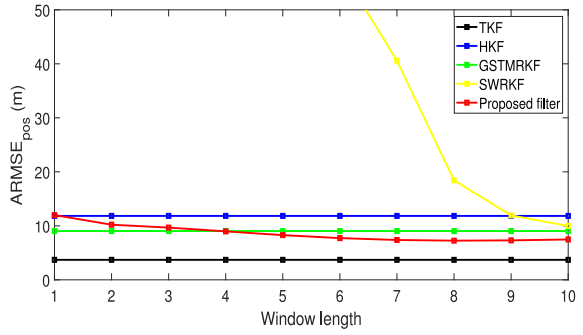


Fig. 7. ARMSE of position when $M = 1, 2, 3, \ldots, 10$.

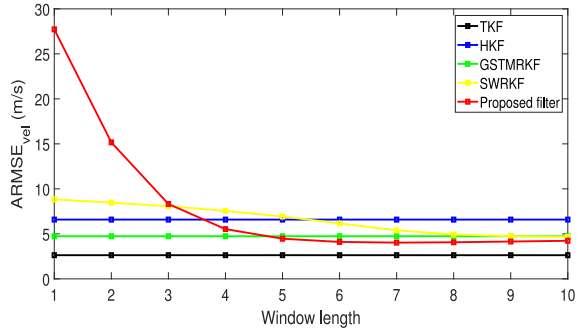


Fig. 8. ARMSE of velocity when $M = 1, 2, 3, \ldots, 10$.

Table 4
The ARMSEs and SSRT of different filters in target tracking.

	TKF	HKF	GSTMRKF	SWRKF	RSTKF	Proposed filter
$ARMSE_{pos}$	3.45	11.11	8.91	84.93	14.11	6.90
$ARMSE_{vel}$	2.39	6.15	4.28	4.25	6.05	3.95
Time(ms)	0.0151	0.0148	0.2110	0.9845	0.4589	0.7049

Figs. 5 and 6 illustrate the ARMSE performances of the estimation obtained from filters with different iterations. It is evident that, across different iterations, the ARMSE value of the proposed filter consistently outperforms that of other comparison filters. In general, different iterations have a minor impact on the accuracy of the estimation. Considering computational complexity, setting the number of iterations to $i = 4$ is reasonable.

ARMSEs under different window lengths M are shown in Figs. 7 and 8. It can be observed that when $M \geq 5$, the proposed filter maintains a stable ARMSE and outperforms the comparative filters in accuracy. This suggests that a window length of $M = 5$ can be considered a turning point in computational efficiency without a significant loss in estimation accuracy. This is because the proposed filter requires a period of measurement information to better identify state and measurement outliers. A shorter M means less measurement information is used in the state estimation, thus reducing the accuracy of the estimates.

Table 5 presents the SSRT of the proposed filter under different M. As M increases, the time required for computation grows almost linearly, indicating that larger window lengths impose higher computational demands. While a smaller M leads to faster computation, it potentially compromises accuracy. Conversely, a larger M enhances accuracy but requires more computational resources. An optimal balance is found at a window length where accuracy is sufficiently high while keeping computational time reasonable. For the proposed filter, an M between 5 and 7 presents an optimal trade-off, providing robustness against outliers and maintaining computational efficiency. In this paper, the window length is set to $M = 6$.

Figs. 9 and 10 illustrate the position and velocity ARMSE under different initial value magnification factors. Specifically, the initial value $X_0 = [0, 0, 1, 1]^{\mathrm{T}}$, and magnified sequentially from 1 to 10 times. The results indicate that the proposed filter consistently outperforms the comparison filter across all initial value magnification factors. Notably,

Table 5
The SSRT of the proposed filter under different window lengths M.

	M = 1	M = 2	M = 3	M = 4	M = 5	M = 6	M = 7	M = 8	M = 9	M = 10
time(ms)	0.1060	0.2021	0.3462	0.4665	0.5945	0.7049	0.8213	0.9506	1.0841	1.2153

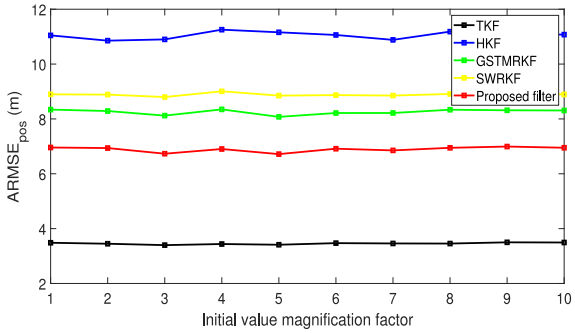


Fig. 9. ARMSE of position for different initial values.

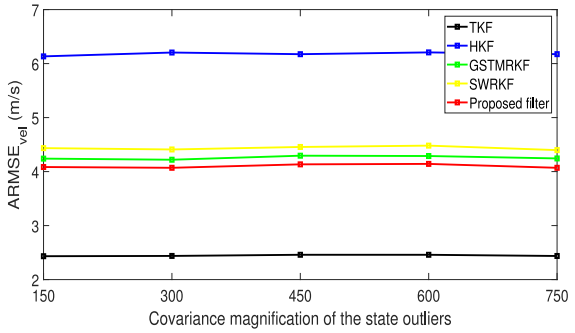


Fig. 12. ARMSE of velocity $U1 = 150, 300, 450, 600, 750$.

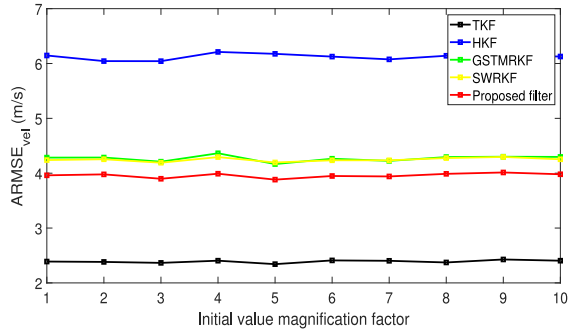


Fig. 10. ARMSE of velocity for different initial values.

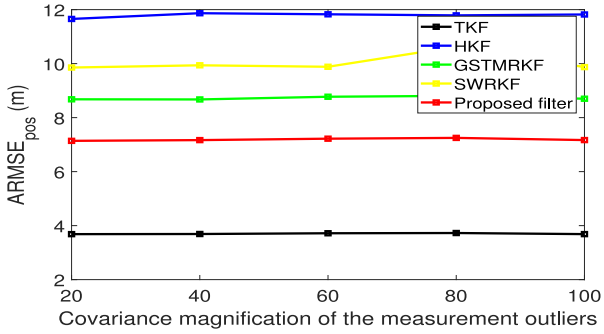


Fig. 13. ARMSE of position when $U2 = 20, 40, 60, 80, 100$.

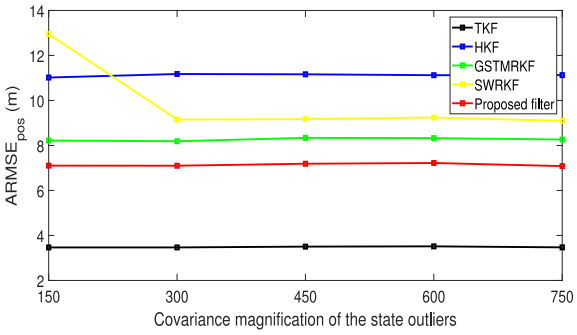


Fig. 11. ARMSE of position when $U1 = 150, 300, 450, 600, 750$.

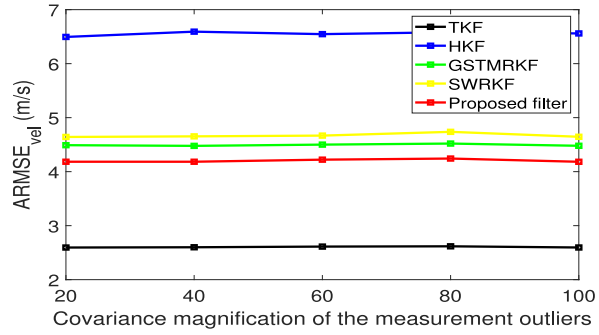


Fig. 14. ARMSE of velocity $U2 = 20, 40, 60, 80, 100$.

the ARMSE of the proposed filter exhibits minimal variation with increasing magnification factors, demonstrating the proposed filter's low sensitivity to initial conditions.

Figs. 11 and 12 display the ARMSE in position and velocity under covariance amplification for different outlier states ($U1 = 150, 300, 450, 600, 750$). The proposed filter demonstrates superior robustness at all U1 levels, achieving the smallest ARMSE compared to the other filters. Figs. 13 and 14 show the ARMSE in position and velocity when addressing measurement outliers with covariance amplification ($U2 = 20, 40, 60, 80, 100$). The proposed filter maintains the smallest ARMSE, confirming its superior performance. This indicates that under varying degrees of noise, the proposed filter still maintains the ability to accurately identify outliers, demonstrating superior robustness.

5. Conclusions

This study addresses the key challenges posed by dynamic outliers in KF through the use of measurement information over a period, namely the effective identification and handling of dynamic outliers. Noise is modeled as a GSTM distribution, and mixed model parameters are jointly inferred using the VB method to enhance the effectiveness and accuracy of the model, which is efficiently integrated into the proposed MHE framework. Through experimental analysis, the optimal window size was precisely determined, further improving the accuracy of estimations and optimizing the algorithm's adaptability to dynamic outliers characteristics. Numerical simulation and target tracking simulation experiments demonstrate the superior performance

of the proposed filter in handling dynamic outliers, showing greater robustness.

CRediT authorship contribution statement

Yue Hu: Writing – review & editing, Writing – original draft, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Weidong Zhou:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no conflict of interest.

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Appendix

A.1. Proof of Proposition 1

Let $\theta_t = X_{t-M:t}$, and utilizing (31) in (30) obtains

$$log q^{(l+1)}(X_{t-M:t}) = -0.5(X_{t-M} - \hat{X}_{t-M|t-M})^{\mathrm{T}} P^{(-1)}_{t-M|t-M}(X_{t-M} - \hat{X}_{t-M|t-M}) - 0.5 \sum_{s=t-M+1}^{t} [(Z_s - H_s X_s)^{\mathrm{T}} \hat{R}^{(l)-1}_{s|t}(Z_s - H_s X_s) + (X_s - F_s X_{s-1})^{\mathrm{T}} \hat{Q}^{(l)-1}_{s|t} \times (X_s - F_s X_{s-1})] + C_{X_{t-M:t}} \tag{78}$$

Applying exponentiation to both sides of Eq. (78) and making use of the Gaussian PDF yields the desired result.

$$q^{(l+1)}(X_{t-M:t}) \propto N(X_{t-M}; \hat{X}_{t-M|t-M}, P_{t-M|t-M}) \prod_{s=t-M+1}^{t} [N(Z_s; H_s X_s, \hat{R}^{(l)}_{s|t}) N(X_s; F_s X_{s-1}, \hat{Q}^{(l)}_{s|t})] \tag{79}$$

According to (79), we can obtain (32).

A.2. Proof of Proposition 2

Exploiting (39)–(42) in (35)–(36), we have

$$log q^{(l+1)}(\eta_s) = \sum_{s=t-M+1}^{t} [(\hat{\sigma}^{(l+1)}_{s|t} - 1) log \eta_s - \hat{p}^{(l+1)}_{s|t} \eta_s] + C_{\eta_s} \tag{80}$$

$$log q^{(l+1)}(\delta_s) = \sum_{s=t-M+1}^{t} [(\hat{q}^{(l+1)}_{s|t} - 1) log \delta_s - \hat{r}^{(l+1)}_{s|t} \delta_s] + C_{\delta_s} \tag{81}$$

Employing (80)–(81) yields

$$q^{(l+1)}(\eta_s) \propto \prod_{s=t-M+1}^{t} [\eta_s^{(\hat{\sigma}^{(l+1)}_{s|t} - 1)} exp(-\hat{p}^{(l+1)}_{s|t} \eta_s)] \tag{82}$$

$$q^{(l+1)}(\delta_s) \propto \prod_{s=t-M+1}^{t} [\delta_s^{(\hat{q}^{(l+1)}_{s|t} - 1)} exp(-\hat{r}^{(l+1)}_{s|t} \delta_s)] \tag{83}$$

According to (82)–(83), we can obtain (37)–(38).

A.3. Proof of Proposition 3

Let $\theta_t = j_s$ and $\theta_t = k_s$, and substituting (31) in (30) gets

$$log q^{(l+1)}(j_s) = \sum_{s=t-M+1}^{t} [j_s d_{s1}{}^{(l+1)} + (1 - j_s) d_{s2}{}^{(l+1)}] + C_{j_s} \tag{84}$$

$$log q^{(l+1)}(k_s) = \sum_{s=t-M+1}^{t} [k_s h_{s1}{}^{(l+1)} + (1 - k_s) h_{s2}{}^{(l+1)}] + C_{k_s} \tag{85}$$

where the parameters $d_{s1}{}^{(l+1)}$, $d_{s2}{}^{(l+1)}$, $h_{s1}{}^{(l+1)}$, and $h_{s2}{}^{(l+1)}$ are given by

$$d_{s1}{}^{(l+1)} = E^{(l)}[log \alpha_s] - 0.5 tr(A_s{}^{(l+1)} Q_t{}^{-1}) \tag{86}$$

$$d_{s2}{}^{(l+1)} = E^{(l)}[log(1 - \alpha_s)] + 0.5 n E^{(l+1)}[log \eta_s] - 0.5 E^{(l+1)}[\eta_s] tr(A_s{}^{(l+1)} Q_t{}^{-1}) \tag{87}$$

$$h_{s1}{}^{(l+1)} = E^{(l)}[log \beta_s] - 0.5 tr(B_s{}^{(l+1)} R_t{}^{-1}) \tag{88}$$

$$h_{s2}{}^{(l+1)} = E^{(l)}[log(1 - \beta_s)] + 0.5 m E^{(l+1)}[log \delta_s] - 0.5 E^{(l+1)}[\delta_s] tr(B_s{}^{(l+1)} R_t{}^{-1}) \tag{89}$$

Using (84)–(85), we have

$$log q^{(l+1)}(j_s) = \sum_{s=t-M+1}^{t} [log[exp(d_{s1}{}^{(l+1)})]^{(j_s)} + log[exp(d_{s2}{}^{(l+1)})]^{(1-j_s)} + log[r_j]^{(j_s)} [r_j]^{(1-j_s)}] \tag{90}$$

$$log q^{(l+1)}(k_s) = \sum_{s=t-M+1}^{t} [log[exp(h_{s1}{}^{(l+1)})]^{(k_s)} + log[exp(h_{s2}{}^{(l+1)})]^{(1-k_s)} + log[r_k]^{(k_s)} [r_k]^{(1-k_s)}] \tag{91}$$

Utilizing (90)–(91), $q^{(l+1)}(j_s)$ and $q^{(l+1)}(k_s)$ are formulated as

$$q^{(l+1)}(j_s) = \prod_{s=t-M+1}^{t} [r_j exp(d_{s1}{}^{(l+1)})^{j_s} r_j exp(d_{s2}{}^{(l+1)})^{(1-j_s)}] \tag{92}$$

$$q^{(l+1)}(k_s) = \prod_{s=t-M+1}^{t} [r_k exp(h_{s1}{}^{(l+1)})^{k_s} r_k exp(h_{s2}{}^{(l+1)})^{(1-k_s)}] \tag{93}$$

According to (92)–(93), we can obtain (47)–(50).

A.4. Proof of Proposition 4

Using (55)–(58) in (51)–(52), we obtain

$$log q^{(l+1)}(\alpha_s) = \sum_{s=t-M+1}^{t} [(\hat{u}^{(l+1)}_{s|t} - 1) log \alpha_s + (\hat{y}^{(l+1)}_{s|t} - 1) log(1 - \alpha_s)] + C_{\alpha_s} \tag{94}$$

$$log q^{(l+1)}(\beta_s) = \sum_{s=t-M+1}^{t} [(\hat{\mu}^{(l+1)}_{s|t} - 1) log \beta_s + (\hat{v}^{(l+1)}_{s|t} - 1) log(1 - \beta_s)] + C_{\beta_s} \tag{95}$$

Exploiting (94)–(95) gives

$$q^{(l+1)}(\alpha_s) \propto \prod_{s=t-M+1}^{t} [\alpha_s^{(\hat{u}^{(l+1)}_{s|t} - 1)} (1 - \alpha_s)^{(\hat{y}^{(l+1)}_{s|t} - 1)}] \tag{96}$$

$$q^{(l+1)}(\beta_s) \propto \prod_{s=t-M+1}^{t} [\beta_s^{(\hat{\mu}^{(l+1)}_{s|t} - 1)} (1 - \beta_s)^{(\hat{v}^{(l+1)}_{s|t} - 1)}] \tag{97}$$

According to (96)–(97), we can obtain (53)–(54).

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